ZERO-TRUST BRIDGE SURVIVAL & AUDIT VERIFICATION ENGINE

Climate-Adaptive Survival Analysis and Paper-Inspection Anomaly Detection

Technical Design, Data, Model Rationale, Validation, and Post-Disaster Triage Report

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Abstract. This report describes a complete classical machine-learning system for predicting climate-driven bridge degradation, auditing reported inspection claims, and prioritising physical non-destructive testing after a flood or cyclone. The system uses right-censored survival analysis, a time-varying Cox model, random survival forests, gradient-boosted survival trees, a robust audit residual, an IsolationForest, and SHAP-based local explanations. The current validated run uses a 1,200-bridge synthetic NBI/IBMS benchmark with Indian district-level monsoon, discharge, and scour shocks. The pipeline is deliberately leakage-safe: reported condition ratings and evaluation-only ground truth never enter the survival feature matrix.

The report is written for engineers, data scientists, and reviewers who need to understand not only what the code does, but why each modelling and validation choice was made.

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1. Executive Summary

The project addresses a narrow but consequential failure mode in bridge management: a structure can look acceptable on a paper inspection while its physical condition, hydraulic exposure, and loading history imply an elevated probability of failure. Conventional bridge management systems often reduce this problem to memoryless Markov transitions between condition states. That formulation cannot represent cumulative scour, monsoon shocks, or the divergence between a reported rating and objective deterioration. This system replaces that view with a right-censored, time-varying survival formulation and a separate audit layer that challenges the human claim.

The validated run trains on 1,200 synthetic bridges drawn from an NBI-style schema across 16 states and 25 Indian districts. The cohort contains 21,691 inspection records, 2,175 district-year climate records, a 52.3% observed failure rate, and a 18.9% latent paper-inspection rate. After bridge-level splitting, the training set contains 780 bridges and 22,154 Andersen-Gill intervals; validation and test contain 179 and 241 bridges respectively.

The Cox time-varying model was selected using validation Uno IPCW concordance, not test performance. It achieved a validation Uno C-index of 0.7719 and a held-out test Uno C-index of 0.7528; Harrell's C-index was 0.7920 and the Integrated Brier Score was 0.0839. The audit layer produced 3 primary paper-inspection suspects and 12 broader alerts on the held-out audit cohort, with an Audit Z-score ROC-AUC of 0.8496.

In the post-disaster demonstration, a compound monsoon event was applied to Darbhanga. Of the 10 test-split bridges in the district, 10 received an immediate NDT recommendation, 8 were classified CRITICAL, and the mean survival drop at five years was 0.521. The mean RUL loss was 13.46 years, with a maximum of 46.17 years. SHAP attributions passed an additivity check with a maximum error of 8.88e-16.

> *Bottom line: the project is operational end to end, reproducible from a fixed seed, and explicit about what is and is not validated. The current evidence is a controlled synthetic benchmark, not a claim of field-certified accuracy on the raw NBI/IBMS extract.*

2. Problem Context and Motivation

2.1 The hydraulic failure pathway

A large share of Indian bridge failures is driven by hydraulic processes rather than by ordinary traffic wear alone. Monsoon rainfall raises discharge and flow velocity; high velocity increases bed and pier scour; repeated flood exceedances remove foundation support; and inadequate waterway area concentrates flow around substructure elements. The consequence is not a smooth linear loss of condition. A bridge may appear stable for years and then lose a large fraction of its substructure capacity in one or two extreme seasons. A model that ignores time-varying hydraulic exposure is therefore structurally mis-specified for this failure mode.

The synthetic generator encodes this asymmetry deliberately: substructure degradation has the highest hydraulic sensitivity, and a separate abrupt shock term is applied when a high-scour year coincides with a flood exceedance. This is why the survival engine includes both smooth age-related deterioration and point-in-time climate covariates.

2.2 The paper-inspection problem

Bridge condition ratings are usually produced by visual inspection. Visual inspection is useful, but it is subjective, intermittent, and vulnerable to optimistic bias. A bridge can be rubber-stamped as healthy while the inspector has not performed physical NDT, has not measured scour, or has not reconciled the claim with the district's rainfall and flood history. The audit problem is not to classify a bridge as safe or unsafe from the rating alone. It is to test whether the reported rating is statistically and physically consistent with objective evidence.

2.3 Why memoryless Markov chains are insufficient

A first-order Markov chain models the next condition state as a function of the current state only. It has no explicit representation of age, cumulative exposure, time-varying covariates, censoring, or the duration already spent in a state. It also assumes that a transition probability estimated from a historical average is adequate for a bridge that has just experienced an extreme flood. Those assumptions are precisely the ones that fail in a monsoon-driven portfolio. Survival analysis is a better fit because it models the hazard of failure over time, supports right-censoring, and allows the hazard to depend on a changing covariate vector X(t).

3. Objectives and Success Criteria

The system is designed around four linked engineering questions. Each question maps to one module and one artefact, so a reviewer can inspect the evidence at each boundary.

Question	Module	Evidence produced
How does failure risk evolve under time-varying monsoon and scour exposure?	1 + 2	Counting-process dataset, survival curves, hazard ratios, C-indices, IBS
Does a reported healthy condition contradict physics and climate history?	3	Physics-implied condition, Audit Z-score, IsolationForest flag, priority
After a disaster, which bridges lost the most remaining life and need NDT now?	4	Post-event survival, RUL, RUL loss, NDT recommendation
Why is a particular bridge flagged?	4 + SHAP	Local additive feature contributions and force plots

Success criteria used in this implementation:

- The survival target must be right-censored and constructed from the minimum of deck, superstructure, and substructure condition.
- No reported condition, latent truth, paper-inspection label, or event time may enter a survival model matrix.
- Train, validation, and test bridges must be disjoint before any imputation or encoding is fitted.
- Performance must be reported with survival-specific metrics, never classification accuracy or standard RMSE.
- The audit layer must produce a holdout finding table and a post-hoc evaluation that is clearly separated from scoring.
- The post-disaster layer must recompute a survival curve, not merely re-rank a static score.
- Every local explanation must be auditable for feature provenance and additive consistency.

4. System Architecture

The architecture is a staged evidence pipeline. Module 1 produces model-ready longitudinal data. Module 2 estimates survival and risk. Module 3 compares the physics view with the human inspection claim. Module 4 updates the physical state after a disaster and converts the new survival curve into an operational NDT priority. The modules communicate through typed dataclasses and persisted artefacts rather than through hidden global state.

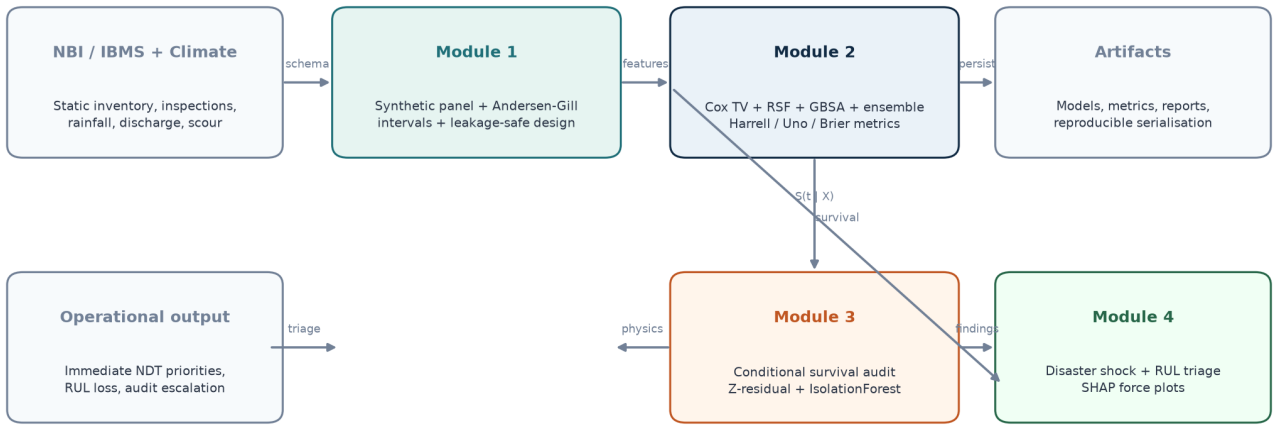


Figure 1. End-to-end module architecture and evidence flow.

4.1 Module boundaries

Module	Primary inputs	Primary outputs	Key guardrail
1. Data pipeline	Synthetic NBI/IBMS registry, district climate panel, inspection history	Static and counting-process splits, sksurv targets, audit report	Bridge-level split before imputation; feature allowlist
2. Survival engine	Preprocessed static and time-varying matrices	Cox TV, RSF, GBSA, ensemble, metrics, hazard ratios	Survival metrics only; test set not used for selection
3. Audit verifier	Survival model, objective bridge metadata, latest climate snapshot, reported condition	Audit findings, Z-score, IsolationForest flag, priority	Reported condition allowed only in audit layer, not survival
4. Triage and XAI	Fitted survival models, post-disaster climate shock, audit findings	Survival drop, RUL loss, NDT priority, SHAP force plots	SHAP features checked against truth/audit prefixes

4.2 The leakage boundary

The single most important architectural decision is the leakage boundary. The survival view sees only objective features that would be available before or at the prediction time: structural attributes, traffic loading, pre-entry climate history, and current climate observations. It never sees the reported condition rating, the latent true condition, the paper-inspection label, the event indicator, the duration, or the inspector bias. The audit view is the only place where the reported rating is an input, because challenging that rating is the audit's purpose. Ground truth is accepted only after scoring, for evaluation.

5. Data Foundation

5.1 What the model is trained on

The current validated run trains on a synthetic longitudinal benchmark generated to match the schema and failure physics of the US National Bridge Inventory (NBI) and an Indian Bridge Management System (IBMS) style portfolio. It is not a raw NBI extract. The generator exists so that the full pipeline can be tested before a multi-gigabyte raw inventory is cleaned, and so that the paper-inspection ground truth is known and controllable for evaluation. The training cohort contains 1,200 bridges, 21,691 inspection records, 2,175 district-year climate records, and 22,154 training intervals.

Cohort attribute	Value
Bridges	1,200
States / districts	16 / 25
Build years	1950 – 2018
Study end year	2024
Climate panel	2,175 district-year rows, 1938 – 2024
Inspection records	21,691
Observed failure rate	52.33%
Latent paper-inspection rate	18.92%
Mean observed duration	28.42 years
Maximum observed duration	70 years
Train / validation / test bridges	780 / 179 / 241
Train / validation / test intervals	22,154 / 5,085 / 6,864

Synthetic NBI / Indian climate benchmark: cohort composition and exposure

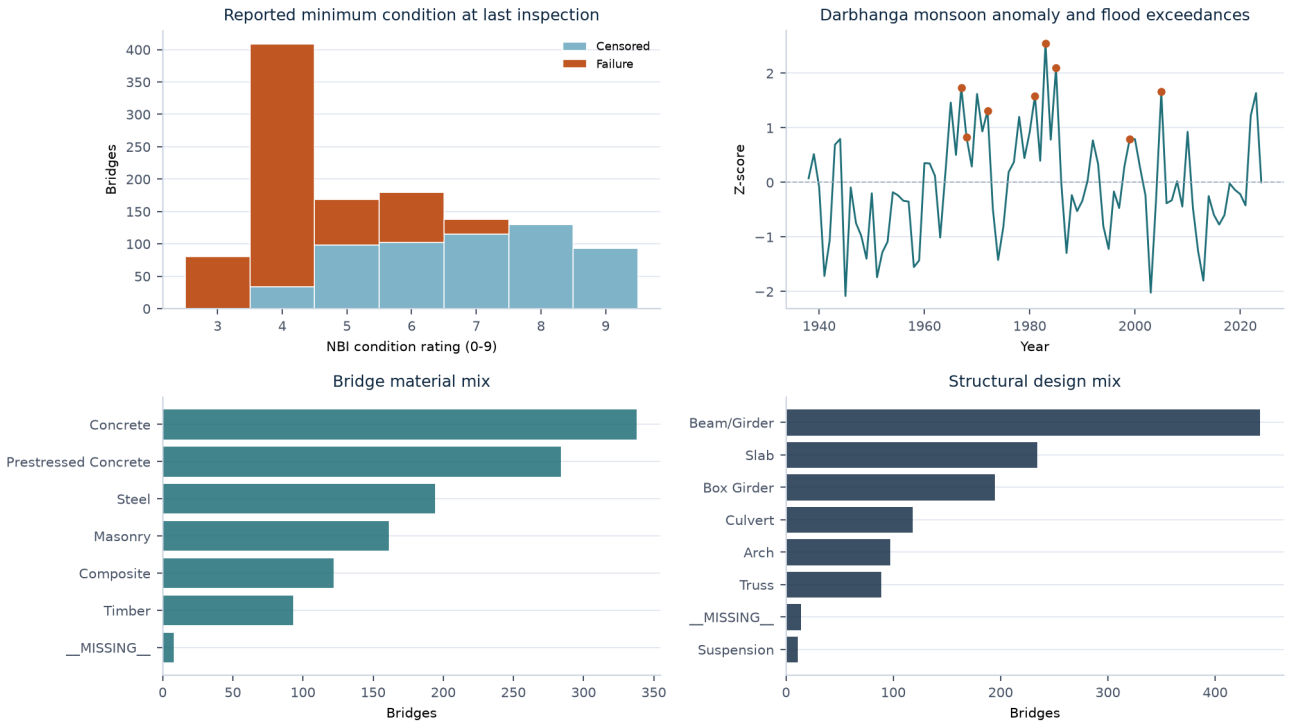


Figure 2. Cohort composition, reported condition distribution, climate exposure, and structural mix.

5.2 Static structural schema

Every bridge has an NBI-style static record. The model uses geography (state, district, latitude, longitude, flood proneness), age (year built), structural form (material, design, number of spans, maximum span, deck length, roadway width), hydraulic context (waterway adequacy, scour-critical flag), loading (ADT, truck percentage, design load, traffic growth), and ownership. Pre-entry climate aggregates are computed from the five years strictly before the bridge enters observation, so the static survival model cannot see future rainfall.

Group	Fields used by the pipeline
Identity and geography	BRIDGE_ID, STATE, DISTRICT, LATITUDE, LONGITUDE, FLOOD_PRONENESS
Structure	YEAR_BUILT, STRUCTURE_KIND_MATERIAL, STRUCTURE_TYPE_DESIGN, MAX_SPAN_LEN, NUM_SPANS, DECK_LEN, ROADWAY_WIDTH
Hydraulics	WATERWAY_ADEQUACY, SCOUR_CRITICAL
Loading and ownership	ADT_TRAFFIC, TRUCK_ADT_PCT, DESIGN_LOAD, TRAFFIC_GROWTH_RATE, OWNER
Pre-entry climate	PRE_ENTRY_MONSOON_MEAN_5Y, PRE_ENTRY_MONSOON_MAX_5Y, PRE_ENTRY_SCOUR_MAX_5Y, PRE_ENTRY_FLOOD_EXCEEDANCES_5Y, PRE_ENTRY_RAINFALL_MEAN_5Y

5.3 Dynamic climate and loading schema

The counting-process view adds year-specific covariates. Age and current traffic are carried forward from the static record. The climate panel supplies a standardized monsoon anomaly, seasonal rainfall, peak discharge, peak velocity, a peak scour index, a binary flood exceedance, and two cumulative counters: cumulative flood exceedances and cumulative scour load. Point-in-time joins use the calendar year at the end of each interval. Cumulative counters are constructed causally and are therefore available at the decision time.

Dynamic field	Definition
AGE_YEARS	Age at the end of the interval
ADT_CURRENT / TRUCK_ADT_PCT_CURRENT	Traffic loading at the interval end
MONSOON_ANOMALY	District monsoon rainfall Z-score relative to the historical reference period
SEASONAL_RAINFALL_MM	Seasonal rainfall total
PEAK_DISCHARGE_M3S / PEAK_VELOCITY_MPS	Hydrological forcing proxies
PEAK_SCOUR_INDEX	Discharge × velocity / 1000, a relative scour forcing index
FLOOD_EXCEEDANCE	Binary indicator that district discharge exceeded its 90th percentile
CUMULATIVE_FLOOD_EXCEEDANCE	Causal count of flood exceedances to the interval end
CUMULATIVE_SCOUR_LOAD	Causal sum of peak scour index to the interval end

5.4 How the synthetic deterioration process works

The generator is not a random-label simulator. It constructs a latent physical condition trajectory for deck, superstructure, and substructure. The annual decrement is driven by a base rate, material and design factors, traffic multiplier, waterway multiplier, span multiplier, a bridge-level frailty term, an age factor, and a climate multiplier. The climate multiplier is exponential in hydraulic exposure, with the substructure receiving the largest sensitivity. An additional abrupt shock is applied to the substructure when high scour coincides with flood exceedance. Reported ratings are generated from the latent condition plus an inspector bias; a configured fraction of bridges receives a large optimistic bias, which creates the paper-inspection signal.

In compact form, the latent condition decrement for component c and bridge i in year t is:

$$\Delta C_{i,c,t} = \text{base}_c \times \text{material}_i \times \text{design}_i \times \text{traffic}_i \times \text{waterway}_i \times \text{span}_i \times \text{frailty}_i \times \text{age}(t)^{0.7} \times \exp(s_c \times \text{hydraulic_exposure}_{i,t}) \times \text{noise}$$

where hydraulic exposure combines positive monsoon anomaly, scour above a threshold, cumulative flood count, and cyclone occurrence. The event is defined when the minimum of deck, superstructure, and substructure latent condition first reaches 4.0 or lower. Inspection timing is discrete: a bridge is observed at one- or two-year intervals, so the recorded event time is the first inspection at or after the latent crossing. If no crossing occurs before the study end, the record is right-censored.

5.5 Climate shock simulation

The climate panel is generated for 25 districts and 87 years (1938–2024). State-level monsoon anomalies follow an AR(1) process with district-level residuals, and known extreme years receive additional boost. Rainfall, cyclone probability, discharge, velocity, scour, and flood exceedance are then generated conditionally. This produces realistic clustering: flood years are not independent coin flips, and high-rainfall years also tend to have high discharge and scour. The result is a benchmark where time-varying hydraulic covariates carry genuine predictive information rather than noise.

5.6 Train, validation, and test design

Splitting is performed at the bridge level before any preprocessing. That prevents the same bridge's longitudinal records from appearing in both training and evaluation. The split is stratified on the event indicator, so each partition contains a comparable failure rate. Validation is used for model selection; test is touched only once for final reporting. The audit verifier is calibrated on the training split and evaluated on validation and test rows. The post-disaster demonstration scores the test split only, because that is the honest holdout view.

Split	Bridges	Intervals	Events	Event rate
Train	780	22,154	408	52.31%
Validation	179	5,085	94	52.51%
Test	241	6,864	126	52.28%

6. Module 1: Longitudinal and Dynamic Climate Pipeline

6.1 Count-process construction

Each bridge record has a duration from entry to the first observed failure or censoring. The Andersen-Gill splitter converts that record into one interval per observed year: (id, t_start, t_stop, event, X_t). The event flag is one only on the final interval of a bridge that failed, and zero on every earlier interval and on all intervals of a censored bridge. This is the canonical counting-process representation used by lifelines.CoxTimeVaryingFitter. The dynamic covariates attached to an interval are the values at the interval end, not future values.

The splitter enforces t_start < t_stop, non-negative duration, no duplicate intervals, contiguous coverage, and an event count that matches the bridge-level target. The pipeline additionally verifies that the count of intervals per bridge equals its integer duration and that events appear only in the final interval.

6.2 Preprocessing and imputation

Missing values are handled with IterativeImputer, which models each incomplete feature as a function of the others. Median imputation is simpler but would destroy relationships between correlated structural and climate variables. The imputer adds missingness indicators so the model can distinguish an imputed value from a genuinely observed one. Categorical fields are one-hot encoded with an explicit missing category, and numeric fields are standardized. The imputer, encoder, and scaler are fitted only on the training split and then applied unchanged to validation and test.

Preprocessing property	Value in validated run
Static input features	24
Static encoded features	91
Time-varying input features	30
Time-varying encoded features	99
Imputer	IterativeImputer, maximum 8 iterations, fitted on training rows
Categorical handling	OneHotEncoder with explicit missing category
Numeric handling	StandardScaler after imputation
Maximum imputer fit rows	50,000 (caps memory on large raw extracts)

6.3 Survival targets

Two target representations are constructed from the same raw records. For scikit-survival, the target is a structured NumPy array built with sksurv.util.Surv.from_arrays(event=..., time=...), with dtype [('event', bool), ('time', float)]. For lifelines, the target is the counting-process frame described above. Having both representations is not redundant: the Cox model needs the interval form, while the tree models need one row per bridge and the censored event time.

6.4 Verification checks

The pipeline executes assertions rather than optional diagnostics. The validated run reports the following checks, all true:

Check	Status
Bridge split disjoint	PASS
Feature allowlist enforced	PASS
Counting process valid	PASS
Events only in final interval	PASS
No future climate join	PASS
No missing after imputation	PASS
Sksurv targets constructed	PASS
Lifelines counting process constructed	PASS

The pre-entry climate check deserves emphasis. The five-year pre-entry aggregates are reconstructed from district records strictly before ENTRY_YEAR. A test verifies the reconstructed mean against the raw panel. This prevents a subtle but fatal leak where a static feature accidentally includes the first observed year or later.

7. Module 2: Survival Engine and Model Selection

7.1 Cox time-varying model

The primary model is a penalised Andersen-Gill Cox model fitted with lifelines.CoxTimeVaryingFitter. The hazard at time t is $h(t \mid X(t)) = h0(t) \exp(\beta^T X(t))$, where X(t) includes the static structural features and the year-specific climate and loading covariates. The model estimates a single global coefficient vector while allowing the covariate vector to change over time. It handles right-censoring and staggered entry naturally, and it produces interpretable hazard ratios.

The fit uses a small elastic-net penalty to stabilise the high-dimensional one-hot design. Near-constant features and highly collinear pairs (absolute correlation above 0.995) are removed before fitting. In the validated run, 99 input features entered the Cox view and 88 were retained. The fitted model converged with an effective penalizer of 0.01.

7.2 Random survival forest

The random survival forest (RSF) is an ensemble of survival trees that split nodes to maximise a log-rank statistic. It can represent non-linear interactions and threshold effects that a single global Cox coefficient cannot, such as a high truck share only being dangerous in combination with poor waterway adequacy and severe scour. The validated run uses 120 trees, log-rank splitting, a minimum leaf size of 3, and square-root feature sampling.

7.3 Gradient-boosted survival analysis

GradientBoostingSurvivalAnalysis fits a stage-wise additive ensemble using the Cox partial likelihood loss. Each stage fits a shallow regression tree to the negative gradient of the loss. This gives a smooth, regularised non-linear risk function. The validated run uses 100 stages, a learning rate of 0.05, depth 3, and 80% subsampling. The shallowness is deliberate: with a few thousand bridges, deep boosted trees would overfit interactions that are not supported by the data.

7.4 Ensemble

The ensemble combines the RSF and gradient-boosting risk scores by rank averaging. Rank averaging avoids the problem that the two models produce risks on different scales. It is appropriate for concordance-based evaluation, where only ordering matters. For pointwise SHAP attribution, however, a batch-dependent rank score is not stable. Module 4 therefore uses a fixed background-calibrated z-score average of the two component risks for SHAP. This is an explicit and documented distinction, not an accidental inconsistency.

7.5 Selection protocol

The four models are compared on the validation split using Uno's IPCW C-index, with Harrell's C-index as a fallback. The test split is never used to choose the model. This matters because the gradient-boosting model happens to have a slightly higher test Uno C-index than the Cox model in this run. Selecting it on test performance would be leakage. The Cox model is selected because it has the highest validation Uno C-index (0.7719) and because its time-varying hazard ratios directly answer the engineering question about monsoon and scour acceleration.

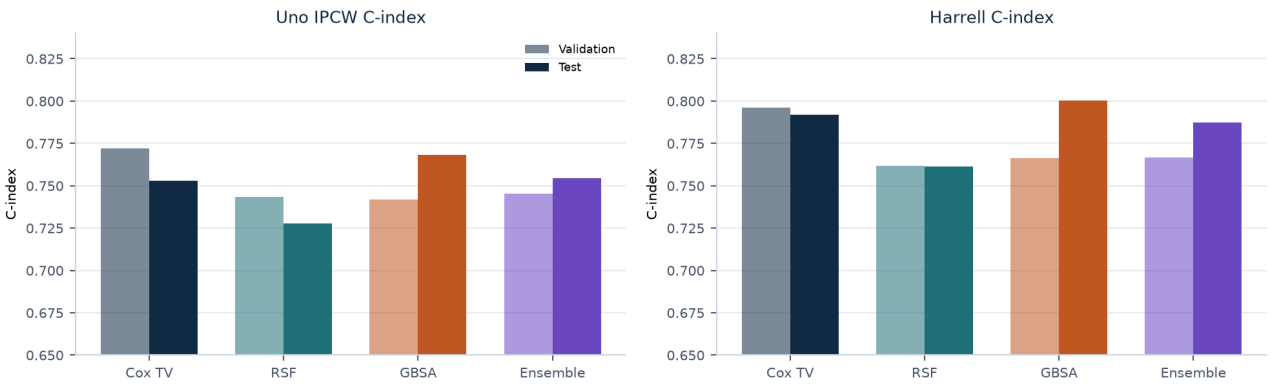


Figure 3. Validation and test discrimination for all four survival models.

Model	Val Harrell	Val Uno	Val IBS	Test Harrell	Test Uno	Test IBS
Cox time-varying	0.7962	0.7719	0.0822	0.7920	0.7528	0.0839
Random survival forest	0.7616	0.7433	0.0763	0.7615	0.7278	0.0830
Gradient boosting	0.7662	0.7417	0.0727	0.8004	0.7684	0.0792
Ensemble	0.7667	0.7454	0.0737	0.7872	0.7545	0.0803

7.6 Survival metrics and why they are used

Harrell's concordance index measures the probability that a model ranks a randomly chosen pair of comparable subjects correctly. It is intuitive but can be biased when censoring is high because it does not reweight censored observations. Uno's IPCW C-index applies inverse probability of censoring weights, making it more appropriate for a portfolio where many bridges are censored at the study end. The Brier score measures the squared error of predicted survival probabilities at a fixed horizon; the Integrated Brier Score integrates that error across the requested horizons. These metrics answer different questions: discrimination versus calibration. Reporting all three prevents a model with good ordering but poor absolute probabilities from looking better than it is.

7.7 What the hazard ratios say

The fitted Cox model is consistent with the intended failure physics. Timber has the highest material hazard ratio in the top coefficient table (2.64 relative to the reference), followed by steel (1.95); prestressed concrete (0.61) and culverts (0.48) are comparatively protective. Truss design has a hazard ratio of 1.61. These are associational estimates from the synthetic benchmark, not causal field estimates, but they confirm that the model is learning the encoded structural priors rather than a spurious label pattern.

Climate shock quantity	Estimated hazard ratio
+2 SD monsoon anomaly	1.0593
One flood exceedance	1.0591
Ten cumulative flood exceedances	7.2802
+40 cumulative scour load	183.0226
Combined extreme monsoon + flood + scour shock	1.1220

The cumulative exposure effects are much larger than the one-year anomaly effects. That is the expected engineering result: a single wet year is less dangerous than a sequence of flood-exceedance years that progressively removes foundation support. It also explains why the post-disaster Module 4 shock adds to the cumulative counters rather than only increasing the

current-year rainfall.

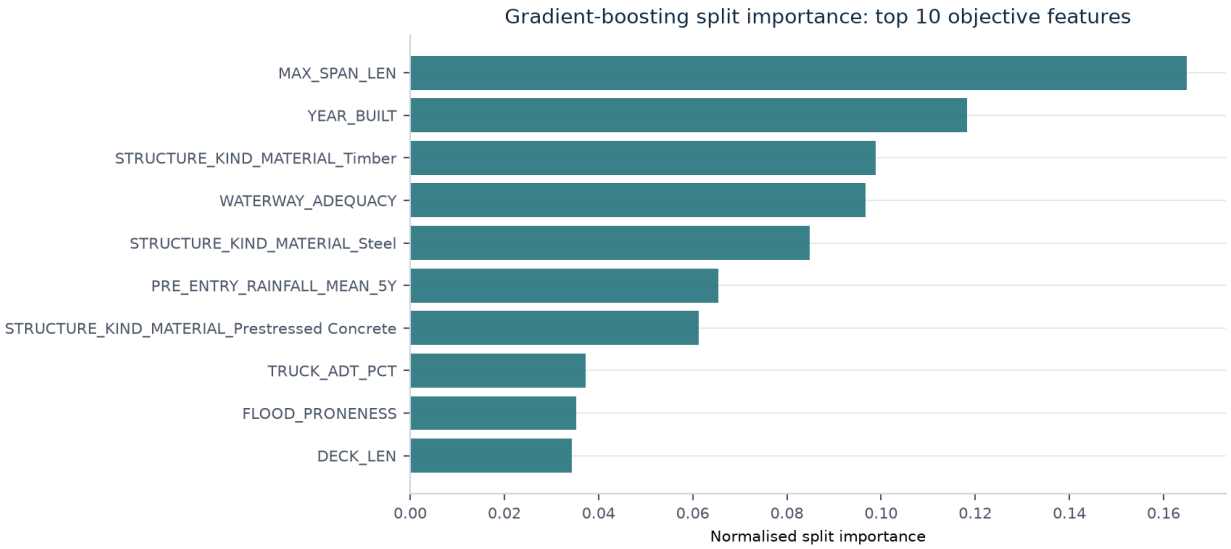


Figure 4. Gradient-boosting feature importance. Span, age, material, waterway adequacy, and pre-entry rainfall dominate the static risk view.

8. Module 3: Paper-Inspection Audit Verifier

8.1 Conditional survival is the correct audit quantity

An old bridge may have a low unconditional survival probability simply because it has already survived a long time. That is not evidence of a false inspection claim. The audit therefore uses conditional survival over the next h years: S(t0 + h | t0, X(t0)) = S(t0 + h) / S(t0). Here t0 is the last observation time and h is the audit horizon, set to five years. The physics-implied condition is the monotone proxy 9 × S(t0 + h | t0), which places survival on the same 0–9 scale as the reported NBI rating.

The primary flag is deliberately explicit: a bridge is a PAPER_INSPECTION_SUSPECT when the inspector reports a healthy condition of 7–8 while the physics-implied conditional survival is below 0.45. The reported condition is used only in the audit table; it never enters the survival model that produces the physics view.

8.2 Two complementary anomaly detectors

The audit uses a robust residual standardiser and an IsolationForest. The residual is reported condition minus physics-implied condition. The centre and scale are the median and 1.4826 × median absolute deviation fitted on the training split, producing a one-sided Audit Z-score. The median/MAD approach is chosen because the residual distribution is heavy-tailed: a handful of paper inspections should not move the calibration. In the validated run the calibration centre was -1.398 and the scale was 1.643 over 780 training rows.

The IsolationForest is fitted on objective climate and structural features plus the physics scores and the reported condition. It therefore flags internally inconsistent audit records, not merely unusual bridges. It is unsupervised because the real world does not provide a reliable label for falsified inspections; the synthetic paper flag is evaluation-only. The audit also defines a broader AUDIT_ALERT that combines the primary flag, the statistical Z-residual rule, and the IsolationForest rule.

Rule	Definition	Purpose
PAPER_INSPECTION_SUSPECT	Reported condition 7–8 AND conditional S(t0+5) < 0.45	High-precision primary finding
Statistical alert	Reported condition ≥ 7 AND condition gap ≥ 0.5 AND Z ≥ 2.0	Catch large optimistic residuals
Unsupervised alert	Reported condition ≥ 7 AND IsolationForest flag AND gap ≥ 0.5	Catch unusual multivariate audit records
AUDIT_ALERT	Any of the above	Broader screening set

8.3 Audit results

On the 420 validation and test bridges, 102 reported a healthy condition and 3 met the primary suspect rule. The broader alert flagged 12 bridges, and the IsolationForest flagged 62. Against the evaluation-only paper-inspection label, the Audit Z-score achieved ROC-AUC 0.8496 and average precision 0.5839. The primary rule had precision 1.000 and recall 0.040; the broader alert had precision 1.000 and recall 0.160.

The low recall of the primary rule is not a coding defect. It is the direct consequence of the stated operating point: a bridge must be both reported healthy (7–8) and below the critical survival threshold. A screening system that flags every old bridge is not useful to an inspection team. The correct interpretation is that the primary rule is a high-precision escalation list, while the broader alert and the continuous Z-score are used for ranking and review. If operational policy prefers recall, the threshold can be raised to 0.55 or the Z-threshold lowered; the report makes that trade-off explicit rather than hiding it.

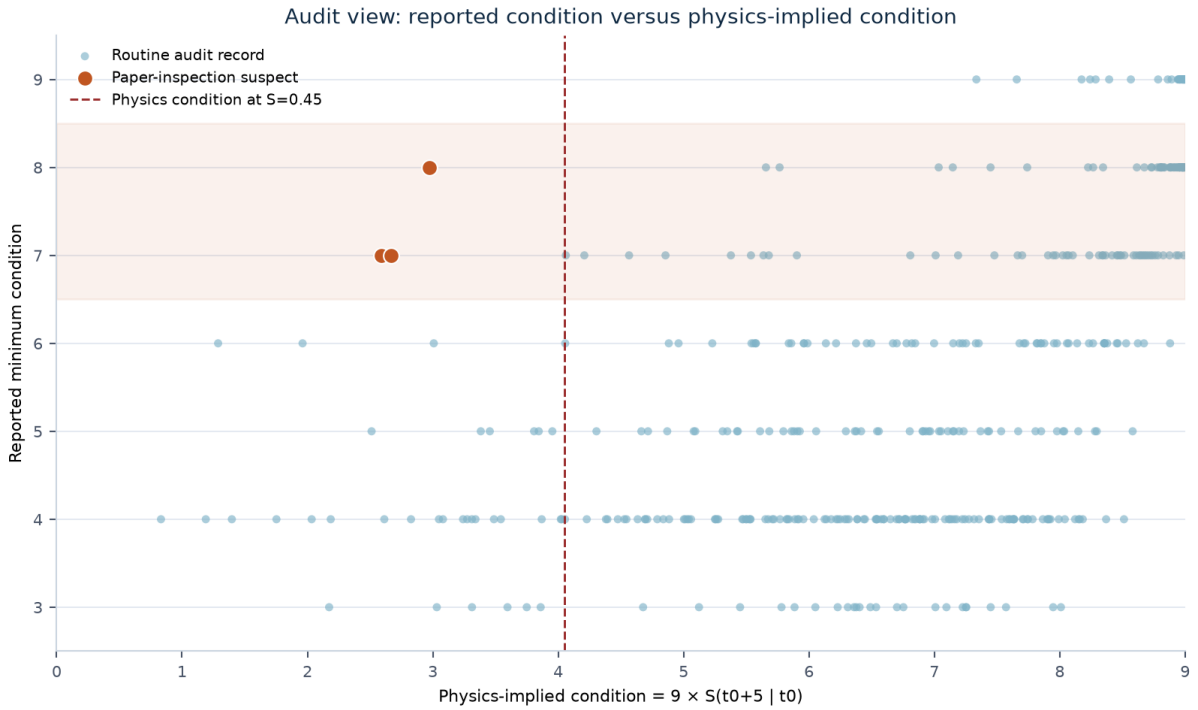


Figure 5. Reported condition against physics-implied condition. The shaded healthy-claim band and the vertical S=0.45 line define the paper-inspection suspect region.

9. Module 4: Post-Disaster Triage and Explainability

9.1 Event representation

A disaster event is a typed object with a target scope (district, state, or explicit bridge IDs), an event type, a severity multiplier, and explicit changes to the climate vector. The event can increase the monsoon anomaly, scale rainfall, discharge, velocity, and scour, set a flood exceedance, and add to the cumulative flood and scour counters. The scope is applied before the shock, so unaffected bridges are returned unchanged. This is important for auditability: the triage output can be reproduced from the event record alone.

The demonstration event is BIHAR-MONSOON-2024, a compound event scoped to Darbhanga with severity 1.35. It adds 3.0 to the monsoon anomaly, multiplies rainfall by 1.9, discharge by 2.2, velocity by 1.6, and scour by 2.4, and adds a flood exceedance plus 15.0 cumulative scour load. These are scenario parameters, not estimates of a specific historical event.

9.2 Survival-curve update

For the Cox view, the updater modifies the last counting-process interval for each targeted bridge. The fitted baseline cumulative hazard and the updated covariate vector are then used to evaluate a conditional survival curve on a one-year grid from zero to 60 years: $S(t_0 + h \mid t_0, X_{shock}) = \exp(-\exp(\beta^T X_{shock}) \times [H_0(t_0 + h) - H_0(t_0)])$. For a tree or ensemble view, the updater modifies the pre-entry and landmark exposure aggregates and the existing step-function conditional-survival helper is evaluated on the same grid. The resulting curve is checked for finiteness, probability bounds, and monotonicity.

9.3 Remaining useful life

RUL is defined as the first time the conditional survival curve reaches the operational threshold $\tau = 0.45$: $RUL = \inf\{h \geq 0 : S(t_0 + h \mid t_0) \leq \tau\}$. The implementation evaluates a discrete grid and linearly interpolates between the last point above τ and the first point below τ. If the curve never crosses τ within the 60-year evaluation window, the RUL is right-censored at 60 years and flagged as censored. This avoids inventing a precise crossing that the data cannot support.

The triage output reports baseline RUL, post-event RUL, RUL loss, survival at the decision horizon, survival drop, priority score, priority class, and an immediate-NDT boolean. The priority score combines current post-event risk (50%), the shock-induced survival increase (30%), and the relative RUL loss (20%). A bridge is CRITICAL if post-event survival is below 0.45 or post-event RUL is at most five years; HIGH if it loses at least five RUL years, has a large survival drop, or is a paper-inspection suspect.

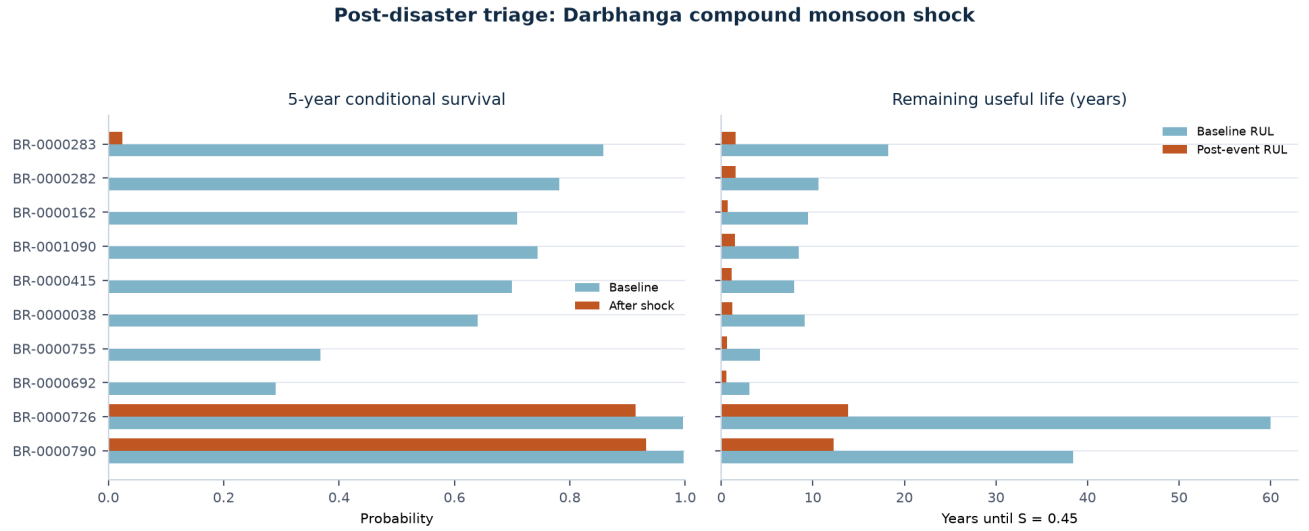


Figure 6. Top post-disaster triage findings: five-year conditional survival and remaining useful life before and after the Darbhanga shock.

9.4 SHAP explainability

SHAP decomposes a model prediction into a baseline value and additive feature contributions: f(x) = φ0 + Σ φj. The project uses shap.Explainer with the permutation algorithm on a background sample drawn from the training cohort. Permutation SHAP is model-agnostic, which matters because the survival models are wrapped objects rather than plain scikit-learn regressors. The explainer is applied to the survival risk prediction, not to the raw survival probability, so positive contributions mean increased risk.

The ensemble requires special handling. SurvivalEnsembleModel.predict_risk rank-averages the two component risks within whatever batch it receives. That is valid for concordance, but it makes pointwise SHAP values depend on the batch composition. Module 4 fixes this by calibrating a z-score for each component on the SHAP background sample and averaging the calibrated risks. The result is a fixed, batch-independent risk function. The report verifies additivity by reconstructing each prediction from its base value plus SHAP contributions.

The validated run explained 8 bridges and wrote 8 standalone HTML force plots. The maximum additivity error was 8.88e-16, essentially machine precision. The contribution table reports feature value, signed SHAP value, absolute magnitude, direction, base value, and model risk prediction for the top ten features per bridge. Audit and truth fields are rejected by the same prefix allowlist used by the survival pipeline.

9.5 Triage results

The event targeted 57 bridges in the district, of which 10 were in the held-out test split. All 10 scored bridges received an immediate NDT recommendation: 8 CRITICAL and 2 HIGH. The mean five-year survival drop was 0.521 and the maximum was 0.834. The mean RUL loss was 13.46 years and the maximum was 46.17 years.

Bridge	District	Baseline S	Post S	RUL before	RUL after	RUL loss	Priority
BR-0000283	Darbhanga	0.858	0.025	18.2	1.6	16.6	CRITICAL
BR-0000282	Darbhanga	0.782	0.002	10.6	1.5	9.0	CRITICAL
BR-0000162	Darbhanga	0.709	0.000	9.5	0.7	8.8	CRITICAL
BR-0001090	Darbhanga	0.744	0.001	8.4	1.5	6.9	CRITICAL
BR-0000415	Darbhanga	0.700	0.000	7.9	1.2	6.8	CRITICAL

10. Why These Models Were Chosen

Model choice is not a matter of using the most complex available estimator. Each choice is tied to the structure of the problem and to the evidence available at training time.

10.1 Why survival analysis instead of classification

The target is a time to a critical condition state, not a binary label at an arbitrary snapshot. Many bridges are still healthy when the study ends; discarding them or treating them as negatives would bias the estimate. Survival analysis uses right-censored observations correctly: a bridge that has survived 20 years without failure contributes information about survival up to 20 years, not a fabricated failure at year 20. It also produces a full survival curve, which is required for conditional audit and post-disaster RUL. A classifier cannot produce a curve without imposing an arbitrary horizon and losing the censoring structure.

10.2 Why Cox time-varying is the primary model

The Andersen-Gill Cox model is the natural baseline for a longitudinal portfolio with time-varying covariates. It supports staggered entry, right-censoring, and repeated intervals, and it produces hazard ratios that engineers and reviewers can interpret. The time-varying form allows the hazard to respond to the current monsoon and scour state rather than forcing the entire effect into a static baseline. Its global linearity is a limitation, which is exactly why the non-linear tree models are also fitted and compared.

10.3 Why random survival forest and gradient boosting are retained

The tree models test whether non-linear interactions matter. In this benchmark they do add value: the gradient-boosting model has the highest held-out Harrell C-index, and the random forest captures span, roadway width, truck share, ADT, flood proneness, and pre-entry rainfall as important splits. They are not selected as the final audit model because the selection rule is fixed on validation and because the Cox model provides the interpretable climate hazard ratios needed for the engineering narrative. Keeping all four models in the report also makes the comparison honest: the selected model is not claimed to be universally superior.

10.4 Why IsolationForest rather than a supervised fraud classifier

A supervised classifier would need a reliable label of falsified inspection. In the field, that label is rarely available: paper inspections are discovered after failures, audits, or whistleblower reports, so the positive class is systematically under-observed. Training a classifier on such labels would reproduce the selection bias. IsolationForest is an unsupervised anomaly detector that asks whether an audit record is unusual in the joint space of objective exposure, physics score, and reported condition. It is complemented by the transparent physics residual, so the final flag is not a black-box anomaly score.

10.5 Why SHAP permutation rather than a bespoke importance heuristic

Global split importance explains the model on average, not why a specific bridge was flagged. SHAP provides local, additive attributions with a consistency guarantee under the permutation formulation. Permutation SHAP is model-agnostic, so the same explanation procedure works for the Cox risk function and the tree ensemble. The additivity check is important: if the base value plus contributions does not reconstruct the prediction, the explanation is not trustworthy. The validated run passes that check to machine precision.

10.6 Why no deep learning

The available cohort is tabular, right-censored, and relatively small by deep-learning standards. A neural survival model would add many parameters, require careful tuning and calibration, and sacrifice the interpretability of hazard ratios, without evidence that the non-linear interactions exceed what survival trees can capture. The stated project constraint is classical ML only. The chosen stack respects that constraint and is appropriate for the sample size and the engineering audit requirement.

11. Evaluation, Validation, and Leakage Controls

The evaluation is intentionally conservative. Model selection uses validation only, the audit calibration uses training only, and the post-disaster demonstration scores the test split only. The following table summarises the headline evidence.

Dimension	Evidence
Survival discrimination	Selected Cox: test Harrell C 0.7920; Uno IPCW C 0.7528
Survival calibration	Selected Cox: test Integrated Brier Score 0.0839
Audit ranking	Audit Z-score ROC-AUC 0.8496; average precision 0.5839
Audit operating point	3 primary suspects; 12 broader alerts
Post-disaster triage	10 immediate NDT; mean RUL loss 13.46 years
Explainability	Additivity error 8.88e-16; 8 force plots
Pipeline integrity	All Module 1 checks pass; all Module 4 checks pass

11.1 Leakage controls

- Bridge-level split before imputation, encoding, calibration, or model fitting.
- Feature allowlists reject EVENT, DURATION, REPORTED_*, TRUE_*, LATENT_*, INSPECTOR_*, and other truth-side prefixes from survival matrices.
- Pre-entry climate aggregates are reconstructed from years strictly before ENTRY_YEAR.
- Point-in-time climate joins use the interval end year, never future years.
- Reported condition is permitted only in the audit feature set, and only the latest minimum rating is allowed.
- Audit ground truth is merged after scoring; it never enters the residual calibration or IsolationForest.
- SHAP feature names are checked against the same truth/audit prefixes.
- The test set is not used for model selection or calibration.

11.2 What the checks do not prove

Passing internal consistency checks does not establish external validity. The synthetic generator is a model of reality, and the paper-inspection labels are simulated. The audit precision and recall therefore measure whether the pipeline recovers the generator's latent mechanism, not whether it would recover real corruption in an IBMS extract. The correct next validation step is a temporal holdout on real NBI/IBMS records with audit outcomes collected independently of the model.

12. Operational Workflow

The system is designed to run as a monthly or post-event batch job. A typical operating cycle is:

1. Ingest. Load the latest inventory, inspection, traffic, and district climate records. Validate required columns and missingness. 2. Score. Apply the frozen preprocessing pipeline and survival model to every active bridge. Compute conditional survival over the audit horizon. 3. Audit. Compare physics-implied condition with reported condition, compute the Audit Z-score, and run IsolationForest. 4. Triage. If a flood or cyclone has occurred, apply the event shock to the affected district, recompute survival curves, and rank RUL loss. 5. Explain. Generate SHAP force plots for the top-N bridges. 6. Dispatch. Send the NDT list to the field team and preserve the model version, event record, and explanation artefacts for audit.

> *The operational output should be read as a prioritisation tool. A high-priority flag means the bridge deserves physical inspection now; it is not a certification that the bridge will fail, nor a legal finding of inspector misconduct. The force plot is a communication aid for the inspection team, not a substitute for engineering judgment.*

13. Reproducibility and Execution

The project is reproducible from a fixed seed. The complete end-to-end command is:

```
python scripts/run_end_to_end.py --config config/default.json
```

A fast smoke run is available with `--quick`. The end-to-end script generates the synthetic cohort, saves the raw and processed data, fits all four survival models, fits the audit verifier, reloads the persisted artefacts from disk, applies the post-disaster event, writes SHAP explanations, and produces a consolidated JSON summary. Reloading the engine and audit artefacts is deliberate: it tests the serialisation boundary, not just the in-memory path.

Command	Purpose
python -m pytest -q	Run all 41 verification tests
python -m ruff check src tests scripts	Static style and import checks
python scripts/generate_report_pdf.py	Regenerate this PDF from current artefacts
python -m bridge_engine.triage_and_xai --help	Module 4 CLI

14. Limitations, Risks, and Honest Caveats

The following limitations are material and should be stated in any deployment review.

- Synthetic evidence. The current results validate the implementation on a synthetic benchmark, not on raw Indian bridge failures. The generator is calibrated to known failure physics, but it is still a simulation.
- Paper-label construction. The paper-inspection label is simulated from an inspector-bias process. Real falsification is under-observed and may have different structure.
- Audit recall. The primary rule is high precision and low recall by design. A deployment must choose an operating threshold consistent with inspection capacity and risk appetite.
- Conditional survival extrapolation. Tree-model survival curves are step functions with a parametric tail beyond the observed domain. The Cox model uses a log-linear tail beyond the largest baseline hazard time. Both are extrapolations.
- Event scenario parameters. The Darbhanga shock is a scenario, not a reconstruction of a specific historical flood. Its multipliers should be calibrated against district gauge and satellite data before operational use.
- Unobserved confounders. Maintenance history, foundation type, riverbed morphology, and inspection quality are not fully represented. Their omission can bias hazard estimates.

- No causal claim. Hazard ratios are associational. They describe the fitted model, not a controlled causal effect of one intervention.
- Distribution shift. A model trained on historical climate may not extrapolate safely to unprecedented rainfall or to a new structural typology without monitoring and recalibration.

15. Recommended Next Steps

- Replace the synthetic adapter with a cleaned raw NBI/IBMS loader while preserving the same schema and leakage checks.
- Add a temporal holdout: train on pre-2015 records and test on later failures to measure forward-shift performance.
- Calibrate district event multipliers against IMD rainfall, CWC discharge, and satellite flood observations.
- Collect independent NDT outcomes and use them to validate or recalibrate the audit operating point.
- Add maintenance interventions as time-varying covariates; the current model treats maintenance as unobserved.
- Add spatial random effects or district frailty terms to account for unobserved local geology and construction practice.
- Add conformal or Bayesian uncertainty bands around RUL and audit scores for operational decision-making.
- Integrate a field mobile workflow so that NDT results feed back into the next scoring cycle.
- Monitor calibration drift and re-fit on a fixed cadence rather than continuously, to preserve auditability.

Appendix A. Selected Field Dictionary

Field	Interpretation
YEAR_BUILT	Construction year; age is derived at the interval end
STRUCTURE_KIND_MATERIAL	Concrete, prestressed concrete, steel, composite, masonry, timber
STRUCTURE_TYPE_DESIGN	Slab, beam/girder, box girder, truss, arch, suspension, culvert
ADT_TRAFFIC / TRUCK_ADT_PCT	Average daily traffic and truck share
MAX_SPAN_LEN	Longest span in metres
WATERWAY_ADEQUACY	NBI-style waterway adequacy rating, higher is better
SCOUR_CRITICAL	Indicator of scour-critical foundation exposure
MONSOON_ANOMALY	District monsoon rainfall Z-score
PEAK_DISCHARGE_M3S	Peak seasonal discharge proxy
PEAK_VELOCITY_MPS	Peak flow velocity proxy
PEAK_SCOUR_INDEX	Discharge × velocity / 1000
FLOOD_EXCEEDANCE	Discharge above the district's reference 90th percentile
CUMULATIVE_FLOOD_EXCEEDANCE	Causal flood count to the interval end
CUMULATIVE_SCOUR_LOAD	Causal scour index sum to the interval end
REPORTED_LAST_MIN_COND	Latest reported minimum condition, audit side only
EXPECTED_SURVIVAL	Conditional survival over the audit horizon
RUL	First time the conditional survival curve crosses the threshold
PAPER_INSPECTION_SUSPECT	Reported healthy condition with physics survival below 0.45

Appendix B. Validated Hyperparameters

Component	Setting
Synthetic cohort	1,200 bridges, seed 42, study end 2024
Split	test 20%, validation 15%
Cox penalty	penalizer 0.01, l1_ratio 0.0
Cox collinearity	correlation threshold 0.995
RSF	120 trees, min leaf 3, max features sqrt
GBSA	100 stages, learning rate 0.05, depth 3, subsample 0.8
Evaluation horizons	5, 10, 15, 25, 40 years
Audit horizon / threshold	5 years / S < 0.45
Audit calibration	median/MAD on 780 training rows
IsolationForest	300 trees, contamination auto
Triage RUL grid	0–60 years, 1-year step; threshold 0.45
SHAP	permutation explainer, background 100, max evaluations 512

Appendix C. Artefact Inventory

Path	Contents
artifacts/data/bridges.parquet	Bridge inventory and last-inspection fields
artifacts/data/climate.parquet	District-year climate and hydraulic panel
artifacts/data/inspections.parquet	Longitudinal inspection records
artifacts/data/static_*.parquet	Model-ready static splits
artifacts/data/counting_*.parquet	Andersen-Gill interval splits
artifacts/data/EVAL_ONLY_ground_truth.parquet	Evaluation-only latent truth and paper flag
artifacts/module2/survival_engine.joblib	Fitted Cox, RSF, GBSA, ensemble
artifacts/module2/survival_report.json	Metrics, hazard ratios, feature importance
artifacts/module3/audit_findings.parquet	Audit scores, flags, and priorities
artifacts/module3/audit_models.joblib	IsolationForest, feature pipeline, calibration
artifacts/module4/triage_findings.parquet	Post-event survival, RUL, and priority
artifacts/module4/xai_contributions.parquet	Local SHAP contributions
artifacts/module4/force_plots/*.html	Standalone local force plots
artifacts/end_to_end_summary.json	Consolidated run summary

Appendix D. Conceptual References and Glossary

The methods used here are established classical techniques. The report does not depend on proprietary algorithms. For deeper reading, consult the original literature on the Cox proportional hazards model, Andersen-Gill counting processes, random survival forests, gradient boosting, IsolationForest, and SHAP. The implementation choices in this project are documented in the module docstrings and in the persisted JSON reports.

Term	Meaning in this project
Right-censoring	The bridge has not failed by the end of observation; we know it survived at least that long.
Counting process	One row per bridge-year with start, stop, event, and time-varying covariates.
Hazard	Instantaneous rate of failure given survival to time t and covariates X(t).
Conditional survival	Probability of surviving the next h years given survival to t0.
RUL	Remaining useful life; years until conditional survival falls to the operational threshold.
Concordance	Probability that a model ranks a comparable pair in the correct order.
IPCW	Inverse probability of censoring weighting, used to reduce censoring bias.
IBS	Integrated Brier Score across a set of time horizons.
Audit Z-score	Robust standardized reported-minus-physics condition gap.
Paper inspection	A reported healthy condition that is inconsistent with objective physics and climate evidence.
SHAP	Shapley additive explanations; local additive feature attribution.

This report was generated from the persisted artifacts of the validated end-to-end run. The seed, cohort size, metrics, and model names in the text are read from those artifacts, so the document can be regenerated after any future run.